1.4.11 Power Set

The set of all subsets of a set A is called the power set of 'A'. It is denoted by $P(A)$.

For example,

(i) If $A=\{2, 3\}$, then find the power set of A.

The subsets of A are $\varnothing$, $\{2\},\{3\},\{2,3\}$.

The power set of A,

$P(A) = \{\varnothing ,\{2\},\{3\},\{2,3\}\}$

(ii) If $A = \{\varnothing , \{\varnothing\}\}$, then the power set of A is $\{ \varnothing , \{\varnothing , \{\varnothing\}\}, \{\varnothing\} , \{\{\varnothing\}\} \}$.

An important property.

We already noted that $n(A) \leq n[P(A)]$. But how big is $P(A)$? Think about this a bit, and see whether you come to the following conclusion:

(i) If $n(A) = m$, then $n[P(A)] = 2^m$

(ii) The number of proper subsets of a set A is $n[P(A)]-1 = 2^m-1$.

Example 1.9 Find the number of subsets and the number of proper subsets of a set $X=\{a, b, c, x, y, z\}$.

Solution Given $X=\{a, b, c, x, y, z\}$.Then, $n(X) =6$

The number of subsets $= n[P(X)] = 2^6 = 64$

The number of proper subsets $= n[P(X)]-1 = 2^6-1$

$= 64 - 1 = 63$

Thinking Corner

Every set has only one improper subset. Verify this fact using any set.

Exercise 1.2

1. Find the cardinal number of the following sets.

(i) $M = \{p, q, r, s, t, u\}$

(ii) $P = \{x : x = 3n+2, n\in \mathbb{W}$ and $x< 15\}$

(iii) $Q = \{y : y = \frac{4}{3n}, n \in \mathbb{N}$ and $2 < n \leq 5\}$

(iv) $R = \{x : x$ is an integers, $x\in \mathbb{Z}$ and $-5 \leq x <5\}$

(v) S = The set of all leap years between 1882 and 1906.

2. Identify the following sets as finite or infinite.

(i) X = The set of all districts in Tamilnadu.

(ii) Y = The set of all straight lines passing through a point.

(iii) $A = \{ x : x \in \mathbb{Z}$ and $x <5\}$

(iv) $B = \{ x : x^2-5x+6 = 0, x \in \mathbb{N} \}$

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